Exercise Sheet 5

Considering the set of tables that we have already seen:

- taxi_stands;
- tracks_braga;
- cont_freguesias

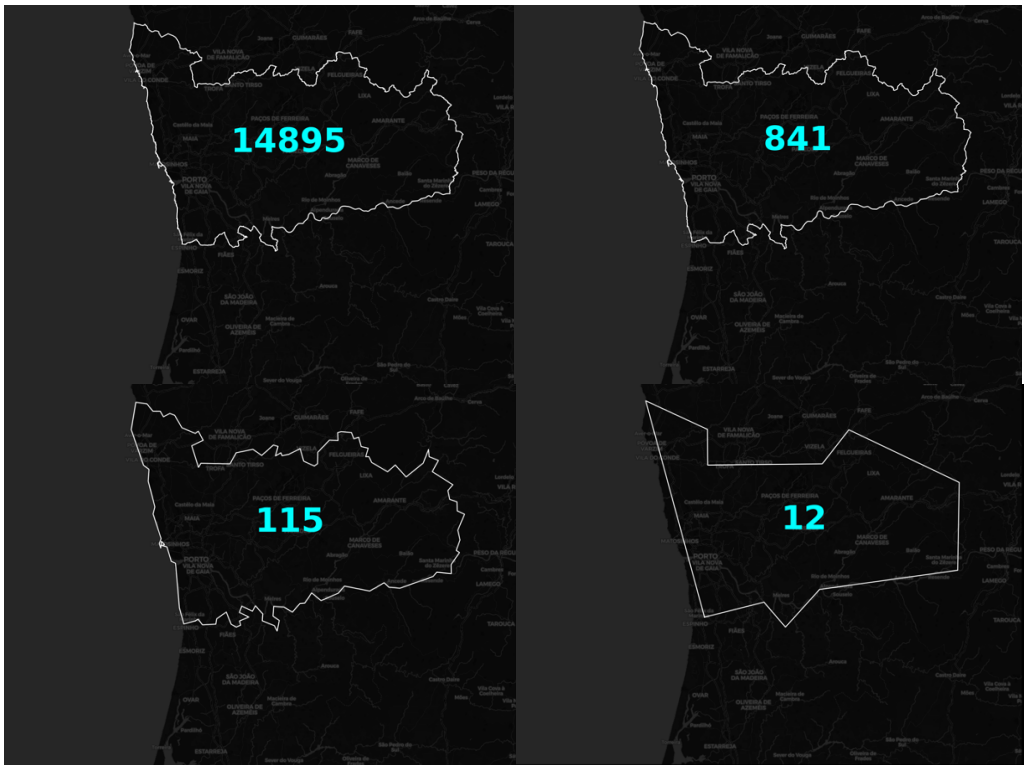
And a new table taxi_services:

```
                                        Table "public.taxi_services"
       Column        |          Type          | Collation | Nullable |                  Default
---------------------+------------------------+-----------+----------+-------------------------------------------
 id                  | integer                |           | not null | nextval('taxi_services_id_seq'::regclass)
 initial_ts          | integer                |           |          |
 final_ts            | integer                |           |          |
 taxi_id             | integer                |           |          |
 initial_point       | geometry(Point,4326)   |           |          |
 final_point         | geometry(Point,4326)   |           |          |
 initial_point_proj  | geometry(Point,3763)   |           |          |
 final_point_proj    | geometry(Point,3763)   |           |          |
Indexes:
    "taxi_services_pkey" PRIMARY KEY, btree (id)
```

use SQL queries and Python with Matplotlib to graphically show what is requested in each query.

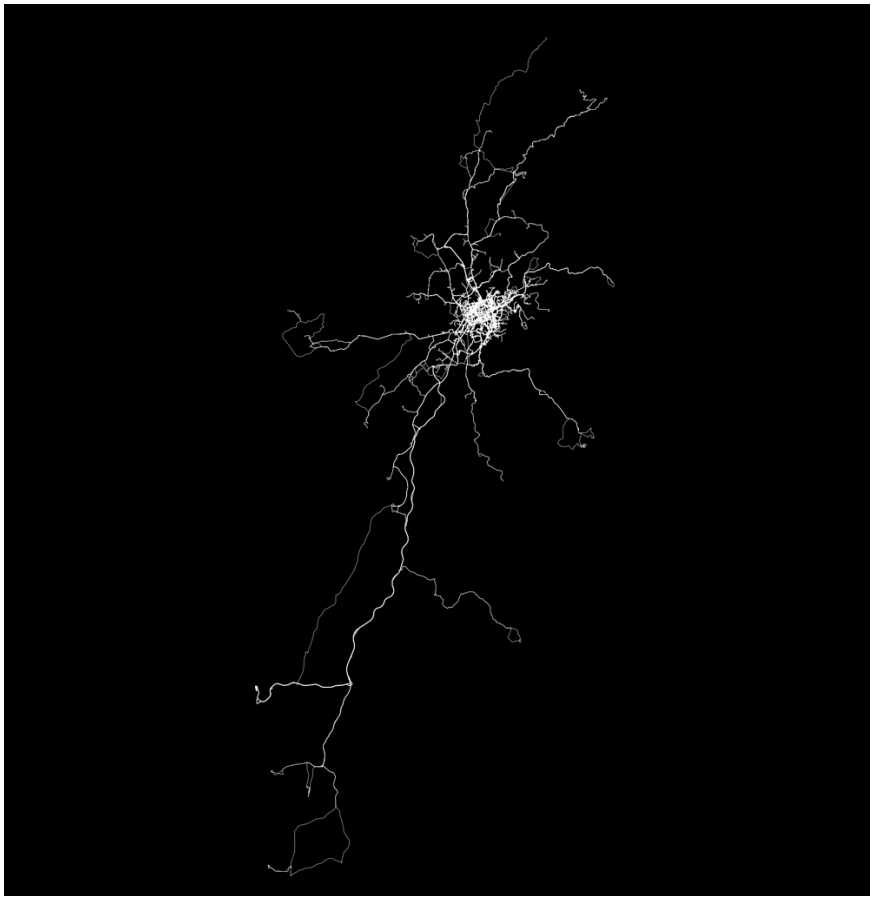
1. Use st_simplify and st_union to create and show a polygon with the boundary of the district of Porto. Draw each polygon with different tolerance in the simplification (no tolerance, 50 meters tolerance, 500 meters tolerance and 5000 meters tolerance). Display the number of points of the polygon as a label in the center of the polygon.

   Example ouput:



2. Display all the tracks from the tracks_braga table, using a slim white line over a black background. Use the zoom lens selector to see the level of detail.

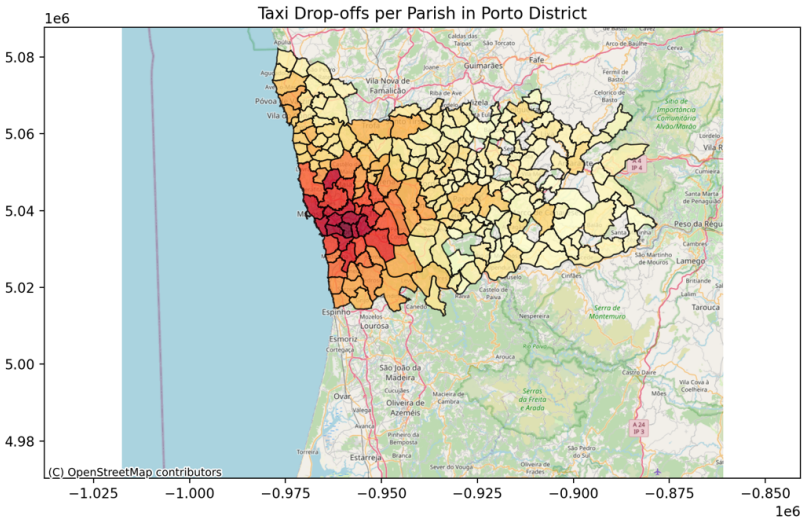
   Example output:



3. Considering the taxi_services table and its final_point attribute, create a map with all the parishes where the service ended. Each parish polygon is colored according to the number of taxi services that ended within its boundaries. A heatmap-based color scale is used, where lighter colors represent fewer services and progressively darker or more intense colors (e.g., orange to red) indicate higher concentrations of drop-offs. This visual gradient allows for quick identification of areas with higher taxi service activity. Use log(count(*)+1) in the SQL query grouped by parish. You can use the following Python code to define the color:

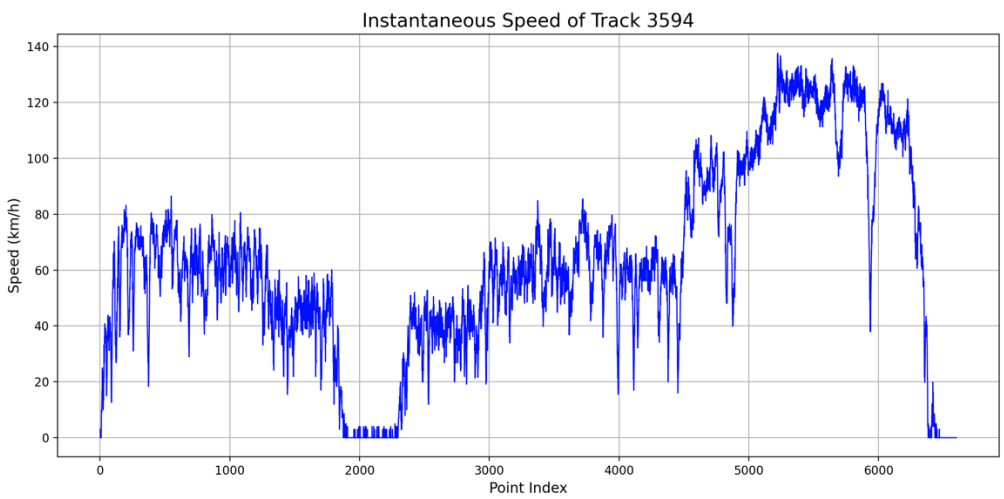
```
norm = plt.Normalize(min(log_counts), max(log_counts))
cmap = cm.get_cmap('YlOrRd')

fig, ax = plt.subplots(figsize=(10, 6))

for geom, value in zip(geoms, log_counts):
    x, y = geom.exterior.xy
    color = cmap(norm(value))
    ax.fill(x, y, alpha=0.8, edgecolor='black', facecolor=color)
```

Example output:



4. Create a function that, given the id of a track, draws a graph of instantaneous speed in km/h. Filter speed errors that correspond to wrong GPS readings. Try with track with id 3594.

Example output:



5. Consider the figure below that lists the tariff rules that are applied in taxis. Create a function given the id of a track and return the value of the taximeter corresponding to that track. If the status is not 'BUSY', it must return 0.

TARIFA URBANA

N.º lugares	Bandeirada		Km	hora	Frações			
	metros	€	€	€	metros	€	seg.	€
4 passageiros								
Tarifa 1								
- diurna	1390	3,25	**0,51**	**16,10**	196,08	0,10	22,36	0,10
- noturna	1112	3,90	**0,61**	**19,30**	163,40	0,10	18,65	0,10
+ 4 passageiros								
Tarifa 1								
- diurna	1390	3,25	**0,65**	**16,10**	153,85	0,10	22,36	0,10
- noturna	1112	3,90	**0,78**	**19,30**	128,21	0,10	18,65	0,10
VEÍCULOS S/ DISTINTIVO								
4 passageiros	1112	3,90	**0,61**	**19,30**	163,93	0,10	18,65	0,10
+ 4 passageiros	1112	3,90	**0,73**	**19,30**	136,99	0,10	18,65	0,10